

CHAWANG, Thailand

WMO: 485570

Lat: 8.424N

Lon: 99.506E

Elev: 100

StdP: 14.64

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	68.6	70.3	64.9	92.4	80.3	66.5	98.0	79.6	10.5	84.7	8.9	88.7	0.0	N/A	0.281

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.1	97.9	78.7	95.6	78.6	94.0	78.8	81.9	91.1	81.4	90.4	80.9	89.7	3.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	154.5	86.3	79.1	151.5	85.8	78.6	149.2	85.4	45.8	91.3	45.2	90.3	44.6	90.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.2	5.5	65.4	99.1	2.2	3.1	63.8	101.3	62.5	103.0	61.3	104.8	59.7	107.0		
(5)			64.8	83.5	2.3	0.9	63.1	84.2	61.8	84.7	60.5	85.2	58.9	85.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	82.2	80.2	81.5	83.3	84.6	83.9	82.9	82.2	82.2	81.9	81.9	81.1	80.3		
	DBStd	2.39	2.17	1.75	2.30	2.50	2.30	1.80	1.71	1.61	1.90	1.78	2.16	1.94		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11743	938	881	1034	1039	1050	987	998	998	957	988	934	939		
	CDD65	6268	473	461	569	589	585	537	533	533	507	523	484	474		
	CDH74	59607	4106	4758	6223	6305	5893	5352	5014	4942	4458	4516	4126	3914		
	CDH80	24274	1595	2295	3195	3094	2466	2108	1841	1842	1574	1561	1362	1341		
(14) Wind	WSAvg	0.7	0.9	1.1	1.0	0.8	0.6	0.8	0.7	0.7	0.7	0.6	0.4	0.7		
(15) Precipitation	PrecAvg	83.5	4.0	2.2	4.0	4.5	7.7	6.4	7.4	8.0	8.3	9.2	12.7	9.3		
	PrecMax	99.7	12.1	9.8	10.7	8.9	10.3	11.7	17.6	12.3	10.5	12.6	21.1	21.1		
	PrecMin	57.7	0.5	0.1	0.0	0.6	5.0	2.7	4.3	5.7	5.2	6.1	6.9	3.3		
	PrecStd	9.4	3.1	2.5	3.3	2.3	1.5	1.9	2.8	1.9	1.6	2.0	4.6	5.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	95.7	99.6	101.7	99.9	94.2	93.5	93.8	93.4	93.2	92.9	92.3		
		MCWB	77.5	76.0	78.5	79.9	80.0	80.6	80.0	79.3	80.3	79.5	79.4	78.6		
	2%	DB	91.3	94.1	97.5	98.4	95.1	92.7	91.7	92.1	91.8	91.4	91.3	90.4		
		MCWB	76.9	76.1	77.8	79.3	81.1	80.0	79.7	79.3	79.7	79.7	79.1	77.9		
	5%	DB	89.9	92.6	95.5	95.5	93.0	91.4	90.4	90.6	90.1	89.7	89.5	88.9		
		MCWB	76.7	76.1	77.4	79.4	80.7	80.0	79.5	79.2	79.4	79.6	79.0	77.6		
	10%	DB	88.2	91.0	93.4	93.2	90.9	89.7	88.7	88.7	88.2	88.0	87.4	87.2		
		MCWB	76.5	76.1	77.5	79.6	80.5	79.8	79.3	79.2	79.1	79.2	78.9	77.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.4	79.9	80.9	82.3	82.6	82.5	81.8	81.8	81.7	81.8	81.2	80.8		
		MCDB	88.9	90.0	92.4	93.7	92.8	91.1	89.6	89.5	90.4	89.6	88.9	88.2		
	2%	WB	79.1	78.7	79.9	81.3	81.8	81.5	80.9	80.8	80.7	80.8	80.4	79.7		
		MCDB	87.4	88.3	91.3	92.6	91.5	89.8	88.7	88.5	89.4	88.7	87.9	87.0		
	5%	WB	78.2	77.9	79.1	80.6	81.2	80.8	80.2	80.1	80.0	80.0	79.8	78.9		
		MCDB	86.0	87.8	90.5	91.9	90.5	88.9	87.8	87.5	88.0	87.6	86.9	86.2		
	10%	WB	77.4	77.2	78.5	79.9	80.6	80.1	79.7	79.5	79.3	79.3	79.1	78.1		
		MCDB	84.7	87.3	89.6	90.5	89.6	87.7	86.9	86.7	86.7	86.5	85.8	84.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	21.1	21.1	19.1	15.8	14.4	13.9	14.3	13.7	13.8	12.9	14.3		
		MCDBR	19.5	23.1	24.7	22.4	18.5	16.7	16.3	17.2	16.4	16.3	16.7	17.5		
		MCWBR	6.4	7.1	7.2	6.2	6.3	5.7	5.9	6.0	5.8	5.9	6.0	6.3		
	5% WB	MCDBR	16.8	19.9	21.1	19.5	16.9	15.3	14.9	14.9	15.3	15.0	14.6	15.2		
		MCWBR	6.5	6.9	6.6	6.4	6.4	6.0	6.1	6.1	6.1	6.1	6.0	6.3		
(40) Clear-Sky Solar Irradiance	taub	0.489	0.498	0.507	0.484	0.458	0.458	0.458	0.459	0.453	0.443	0.454	0.472	0.482		
	taud	2.161	2.122	2.133	2.201	2.311	2.344	2.326	2.336	2.357	2.321	2.253	2.199			
	Ebn at Noon	262	264	262	265	266	263	263	269	276	273	265	261			
	Edh at Noon	48	51	51	48	42	40	41	41	41	42	44	46			
(44) All-Sky Solar Radiation	RadAvg	1421	1743	1796	1765	1552	1505	1467	1507	1474	1353	1223	1216			
	RadStd	134	134	170	103	89	83	82	81	49	108	100	119			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (17 neighbors)		+0.32	+0.49	N/A	N/A	+0.32	+0.36	N/A	N/A	+110	+110

Nomenclature: See separate page